

I led the development of the project's core codebase, taking primary responsibility for the design, implementation, and integration of the computational components. In parallel, I managed the GitHub repository, including version control, branch management, and the resolution of merge conflicts, ensuring a stable and well-documented development workflow throughout the project.

A significant part of my contribution involved the formulation and justification of quantitative performance metrics used to evaluate the model. I defined these metrics in line with the mathematical and financial objectives of the coursework and provided a rigorous mathematical interpretation of the models' behavior and outputs, linking theoretical assumptions to empirical results.

I was also responsible for hyperparameter tuning, working closely with team members who focused on the economic and financial analysis to ensure that model calibration was both mathematically sound and economically meaningful. This collaboration helped align computational performance with financial interpretability.

In addition, I introduced the team to LaTeX for professional mathematical and technical writing. I took responsibility for assembling the individual contributions into a single coherent document, ensuring consistency in notation, structure, and presentation, and producing a final report that met academic standards in mathematical finance.